

Active InferAnt Stream 012.2 ~ Case of the Evolving CEREBRUM: The Curious Language of the Incident

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All right. Hello, welcome. It is April 16th, 2025. This is active in France stream 12.2 to and we will begin it with making a GitHub push to the repo that all of the work is going to be in continuing on from 12.1 start stream. This is going to the GitHub repo active inference institute/ cerebrum and looking forward to people's ideas and questions during this stream. So this is the case of the evolving cerebrum, the curious language of the incident. We're going to be using cursor IDE version 48.9 on the native edge of development. LLMs. The two main LLMs that are being used uh in this development have been Claude 3.7 and Gemini 2.5 Pro EXP 325. So I'm just going to delete things as we go through them and look at some practical and some theoretical aspects of cerebrum. All right. So uh to begin with and sort of to catch up on the zenotto um the version one was uploaded on April 7th and commonalities similar topic. Also, there was a uh script that was in an older folder called cerebrum in the repo that was used to assemble this the the paper. Um, in doing that, learned a lot about assembling papers and writing them, expanding them. And in this current version 1.4 release today, it's up to 146 pages and still generated by a single simple script, which we'll look at in a second. um and how it's expanded. So there's this sort of paper component and the paper's generated from the paper folder in the repo. Let's look into it in cursor first how the paper is made. Then we will look at some cool applied outcomes like making a linear regression neural network and partially observable marov decision process model declining those into different cases and getting different uh kinds of behavior from different from the same model but of different model types and getting a little bit clearer of a view in a few specific cas. cases what it means to have a given generative model in a different case. Okay, first the paper then to the source code and um deleting things along the way adding things as people add anything in the chat. All right, first the paper generation. Oh, and the the documents assem uh the read me in the in the paper folder describes this process. Assemble paper is a script of 551 lines that looks into the component section and assembles them into the output. So in the output there is the PDF

that's generated which concatenates the title page makes a table of content including all the main text sections however many there are and then all of the supplements however many there are. Um, it also outputs a pure markdown concatenated version of the assembly process and it outputs an HTML version. So here's a website rendering clickable links still has the inserted images. So a lot of flexibility in how and also the the paper and the images are also vastly improved. So there's multiple ways to make static documents, interactive websites, um, and markdown files for easy machine reading and formatting as well. It's generated from the components folders. There's templates so you can add more sections. Then there's three folders, figures, main text, and supplements. So in the figures are the 15 current figures and they have their name and title. There's there's more general ways to do it uh using LaTeX overleaf all these kinds of things so that the numbering is automatic as well. So this is kind of a fixed numbering system but could be generalized too. So, it has for each figure a caption, a title, a caption, and the mermaid markdown code to produce the figures. Those get saved into the output as direct PNGs. And also, there's a graphical abstract, which looks a little better in PNG, which has the figures and the abstract. So kind of like scientific poster style. Then there are the main text and the supplemental information sections. The main text is again determined by the numbering of the file. But these are markdown files with no mermaid diagrams but citing the mermaid diagram for insertion and linking. So these are markdown files in the main text. title page, abstract, introduction methods, case functions and cognitive models, model workflows as case transformations, active inference models and conclusion. Then there are the supplemental sections and if uh people have ideas for other supplemental uh sections to add, we can just add a 13, 14, etc. Um, I'll open up the website just to get a look at what's in the supplements. Supplement one is a mathematical details and glossary. Supplement two breaks out the novel proposed linguistic cases. Supplement three focuses on different practical applications. Four is a bibliography type section with related work. Five goes into more category theory details. Six gives future directions and road map. Seven goes into some computational complexity estimates for different transformations. Eight goes into more detail on active inference. Nine, more information on math. 10, algorithmic details. 11, the formal definitions of the core linguistic cases, the main ones that are focused on in the paper. And section 12, uh, case study on writing a research paper and how in different sections of the paper, there's different ways that sections are used and do different things with respect to each other and the ideas and the reader. So that different cases could be applied to characterize the nature of a model or a process in that way for those phases. Ju just going through this last one

just as an example of a supplement and kind of thing that if we wanted to build another supplement someone can make a suggestion. Precision management in research paper writing equivalent to learning rate, forgetting rate, um attention, precision broadly just describing modulatory strategies for different kinds of upweightings and down weightings. Message passing occurring amongst author and manuscript, amongst sections, manuscript and readers and reviewers with kinds of intent oriented um entities in different cases or using a similar joint model of communication in in different settings motivating why to have cases. writing strategies. Haven't looked into any of these. Maybe some are useful. Um, how to make certain kinds of pivots rhetorically in, I guess, speech or writing and different ways to interpret how uh regimes of attention change with with uh across different sections. All right. So that's the paper on Zenoto and we can we can add more just to close off this section on the paper assembly process. There are probably other tools to to do this more flexibly. There's things like deep writer and a lot of other approaches to doing structured document generation. Building it from scratch uh with cursor is still pretty interesting. And having uh plain text, multiple sections, possibly doing um lumping and splitting like have introduction 3.1, 3.2, 3.3, or just a more flexible way to order the sections. It makes it really easy to go into a section and add some kind of feature using cursor. Like here in the introduction, just add one relevant paragraph to the end. Structured smaller forms. figuring out the right kinds of core graining. Obviously, just having a dictionary isn't writing a paper. So, having things broken out to the word level might be kind of a granular way, but maybe there's something interesting about that format. Having a lookup table of a dictionary and encoding sections and words in in a totally graphical way. just easier to write and version. Just small updates to the text can can be done and reviewed. So, collaborative editing can can zoom in and and break out different sections. it'd be pretty clear um if there's a flexible ordering method or just with the file names even if someone broke out 4 A and 4B or 4.1 4.2 updated the assembly script etc. So okay that's the paper. Um next uh how about the examples of the specific models then the documents then anything that that people have brought up by then uh bigger picture how do we integrate case-based logic so nouns in different roles and sentences. However, treating models as sort of noun entities, how do we do that in theory and practice? Different ways to make visuals. Mermaid's been super useful. Maybe there's other plain text formats that are useful too, but creating good system and process diagrams, structuring information from pros and tables and lists math with latte. All right. Linear regression pom dp neural networks. So this is in the tests in in the source src folder. tests subfolder. Let's see if code viz will give a

good representation of this. Okay, maybe not today. Uh in the tests folder, there are uh several script files testing each of these different models, some utilities, and a readme. Um the three that are fleshed out at this point are neural network, linear regression, active inference, POMDP. So in each of these cases, there's going to be a single model that gets defined and then it's going to be uh deployed in different ways leading to visualizable analyzable differences in in outcome. So let's just look at some of the images first. Okay, here's a graphical summary of how for the very same neural network, whatever kind of of technique is being used, there are different ways that just considering neural network like a noun, let alone the reference of what it means. But there are ways that people talk about models. That's the curious language is the what people talk about when they talk about models. And it's right there in the sentence though in a way silent in most cases in English. There's the neural network does something standard dictionary form nominative. The researcher trains the neural network like the researcher looks at the neural network accusative case recipient of action. Direct and indirect similar with dative more of like to give rather than to look upon genitive outputs of one of the cases that's visible in English with the apostrophe s like the tables leg. So it's the outputs of the table and these are uh it's a different way to talk about the outputs of something just at the first pass. It's used in a different way in the sentence. And then looking from the way it's used in a sentence on up through what the real reference is, it starts to become a little clearer how these kinds of sentiments that are already expressed in natural language in method sections in people's own inner voice and conversations around modeling, different ways that the words are used correlate or relate to in this linguistic fashion to differences in the deployment of the model. Hence possibly regularities that can be described and also uh structured in ways that are outside of these. Okay, just the last cases here instrumental by means of the team solves a problem by using or with a neural network. Locative the patterns exist within the neural networks layer. So referring to the model as a location where there are spaces within or being in time within the model. Ablative the errors originate from the neural network weight. It's kind of an interesting case. Um has some similarity with genative in some situations. And then the vocative which is the direct addressment like calling a different digital assistant with a name or calling the model to see if it exists like a ping. Then in each of those cases the same neural network gets uh run with these slight differences. So it's it's it's pretty variable per um okay so in the ablative case it was discussing the errors of the model. So in the accusative the model was being trained. So this is just a trace a as would be in some other kind of machine learning paper describing the target of

optimization like in the method section where we say we train the neural network to dot dot dot. here in the dative was the giving data. So this is just describing as a function of input distribution what do different activations look like in the model. So as it's being given different gifts what patterns are arising same neural network here uh in the genitive case is describing uh function outputs output distributions Some animations of generating output instrumental. This might be showing different ways that statisticians do things with the model. Looking at inputs, outputs, activations. Looking at the structure of the model. Looking at the model's training history for a given curriculum as a kind of empirical data. Looking at kinds of interpretabilities, looking at the structure of the neural network as as something that can be done with like tool engineering. All right. Locative describing possible locations inside of not not super fleshed showing the model in its agentic operation. Like when people say the model does this, the model learns to fit this working as active agent generating predictions. Though those predictions would be the predictions of so it could be the same reference could be used in in within one sentence A threw the ball to B and then B threw the ball to A. So within one sentence or across sentences different nouns are used in different cases and they're marked differently from a written from a speaking perspective in different languages. Similarly, models as like nouns are used in sentences in different case settings. So those have different implications for what is viewed, what is meant, what kind of capacities would be required from that model. Like does the model need to turn on? Does it need to just give a ping like with a vocative or does it need to make a prediction? So that's the neural network example. Now there's two more examples. There's the linear regression. Next, so here is a $y = mx + b$ type linear regression. So ablative predictions of the model there were some very interesting diagrams that were arising in this linear case. So here are talking about the model as a causal originator. So there's the model which causes predictions which influence or lead to decisions which have these sort of causal consequences. So that's model as a source of of causation and prediction cross validation here there's an evaluator who is eval using doing something such that they it is said that they evaluate that model so that the model is being here evaluated to well residuals analysis doesn't have anything. Cross validation in the evaluation phase. Here the model is in the dative case the recipient of data like I sent I gave a present to my friend. I sent data to the model and that is reflected in the batch ingress process with different sets of data um or like perfective and imperfective like the streaming online real-time data model was receiving data in a continuous and then the perfective like The model finished receiving 50 units of data. Genative uh is like apostrophe s. So it's the possessives of the model. So it's the

intercept of the model, the slope of the model, mean squared error of the model. All these kinds of statistical expressions, things that might be in a results [Music] section, instrumental used as a instrument of action, used as a tool like we hit the nail with the hammer, we analyze the data with the model. So it's doing different again variable functional uh images but looking at different kinds okay so we trained this model best as we could whether it was this or that method and then now we're going to look at the correlation or the causal interaction between factors one and two and then factors one and three and then we're going to use that instrumentally and say oh okay here's a 0.14 correlation and and we're this confidence this far away from zero and here it's this correlation. So this is these two features are more correlated or they have this causal relationship. So doing things with models locative positions within the model. So where is the model located in information spaces? That's that's one that's the kind of interpretation of the locative and and there are multiple um senses in which a given case is used. But one of them was highlighted in that neural network example which was a location in the model like on the third floor of the neural network. And here that would still apply. You still could talk about I'm talking about the y in the equation. And then here's another view which is the location of the model like they are at the store. The model is at 2, 3. Nominative. This is sort of even though the the uh animation doesn't really work. The nominative is when using the standard dictionary type sense of the word when the model does something like that model fit the data that entity did that model as actor fitting the data. model fitting data. All the kinds of sentences where the model is it it could be replaced like with a person's name or a doer vocative. So here this has some uh there's a few funny ways the vocative is used. This one is in the vocative case data points directly address the model. So uh each point communicates its residual value to the model line. This direct communication visualizes the conversation between data and models similar to how we might address someone by name. So this is almost at a sort of algorithm as communication level thinking about information processes within like oh -2.55 told the slope where it was and the slope told back its residual. So that kind of whereas in the neural network case the vocative focused on the addressment by name for like a digital assistant. Okay, here's some other vocative uses even in linear regression. So, it's the data and then the the model's prediction and then is saying I I'm telling you how far I am under the value. Another interpretation of algorithm as communication where a given variable X is saying hey slope multiply me. So $x * 1.97$ and then to the intercept the other part of the $mx + b$ says add 4.98 and then combine us both or we get combined or I'm the final answer after combining. It's sort of like a linguistic approach to describing

information flows and algorithms. Okay, so that's the uh linear regression model and this is all from like test linear regression test neural network test active inference palm dp. So here in the in the palm dp case um for a given model it's being described in different cases. So just to read the sentences, I'll just say the model instead of the pom dp. The model decides which action to take next. Nominative. The programmer evaluates the model's performance. Accusative. The environment gives observations to the model. Dative the model's belief about the state guides decision. Genative. We solve the problem using a model framework. Instrumental. Different policies exist in the model state space locative. The new algorithm deviates from the standard model approach ablative and hey model what action should I take next focative. So now let's look through those. Start at the bottom. So here is a direct addressment like the sort of digital assistant addressing a model and and also which kinds of functions are going to come into play. So this gets towards uh one of the use cases for having this type of lexical intelligence for modeling is if one expression block or methods block has a bunch of vocatives then you know that for whichever agents are addressed these are the only commands that you need to spin up. Whereas if it was some other case like it making a prediction or like you know doing forward inference might be a different computational setup than getting trained than getting evaluated. So in that way just you can say like what was happening with one actor through a paragraph where they were being referred to in different cases you can trace the same model as it's being referred to within and across sentences in perhaps different ways in different computational setups and and and build for that. Okay. So here is focusing on the vocative nature of the call and response between the agent and the environment in the nominative. So here's the model in determination of what to do for example by doing policy inference like expected free energy policy selection as inference on which actions I select given the states and those might invoke policy inference methods simple 2x2 matrices for the observation transitions. So this is tracing out the decision making process looking at the palm DP as an agent and as it's learning and updating its belief. So what and just here just an example animation what is its belief and action conditional upon observation. Okay, in the locative here is using again the um a little bit of a fusion of referring to policies existing in the state space which is pointing to the models state space. So in that way referring to like a location in the model. However, the location of the mo of of the the location of the state space in the model may be locational. So like the location parameter of the model is in 3, 4. So here referring to the point the phase space location. So all the kinds of theory and empirical aspects of saying like the the model oscillated around a point attractor or the the model made its way from

thinking it was a cool day to thinking it was a warm day. Those are paths and passages in phase space. So those may use verbs of motion or locative in a certain way. instrumental people saying stuff like we solve the problem using the model. So showing the model looking at different ways in which the model is used to explore questions. So making predictions, looking at phenot computational phenotypes like the magnitude of belief updating or the the um different aspects of the simulated or actual behavior of of model being being used as a tool. Looking at different statistical traces from the tool. looking at error. Um, envelopes generative is like the the the genative AI the generative AI like the output of GPT or the text wrote GPT's text the text wrote by GPT predictions of the model the model's predictions Looking at the beliefs of the model. Looking at the collection of them. Looking at the predictions of the model. Looking at the beliefs of the model in relation to observations of the model. Or it could be said in relation to observations sent to the model. So it' be like another way to use observation and basically convey the same syntax which is like we all get it. It's getting data points but literally as it would be with a noun even though the two expressions the obser as a function of observations of the model as a function of the model's observations is using model in the genative as having observations that are possessed by it or you could focus on the the sender of the data and say that the data were sent to or the observations were sent or x sent object observations to the model. So those two features might both be true. They might be isomorphic. They might highlight different features of that setting. But that's just an example uh of how many ways there are to map expressions about models, model linguistics to models in in theory and practice. Just like there's a ton of stuff that could be said about a tree and just the tree grows, the leaves of the tree, people looking at the tree, like all those things that are kind of kicked off by prepositions and context. All right, last ones for the uh POM DP and then to the documents and then to any uh ideas or questions people have. here in the native environment giving observations to the pom dp like what it's direct indirect object the pom dp gets observations from the environment is having the pom dp doing something getting and then here the environment gives observations to the pom dp so looking at the impact of observations which might describe very well the same graphic or or related graphic as was used. Just like the mass of the leaves and the branches mass are referring to different levels of analysis of a tree and they're using a different case, but they still may be very related measurements. And more importantly, we're talking about the same tree. So this is the same POM DP, possibly the same run of the same POM DP. But if you start to think about just models as they are spoken about the kinds of things people say about models like first we

designed the model then we use the model to generate this then we sent training data to the model we got the model's errors as it's like that kind of talking about models aspect sometimes refers to the very same model in the exact same situation other times it refers to the same lineage of model being used in different ways. Like if it says we used the model to sweep across this parameter range or parameter range was swept across those kinds of things but then maybe only some of those parameter uh parameterized models were taken forward and then they were used to generate predictions. So just it it could be a variant of something that was described like not exactly the same single file. But in these simple examples, it is like the exact same generative model. All right. So that's the uh a bit about the testing suite and uh test driven development is very effective in this kind of augmented coding setting. It can help lock in a lot of functions um a as can good use of git. Okay. Now to the doc section which actually has many many many documents. So uh we will definitely not look at all them but at the top level of the docs folder there's some information about cerebrum just how it works this live stream file um different features about cerebrum maybe these will go into a folder and then there's several folders so starting from the smaller ones uh dialogues these are like 17 different genres of sort of creative dialogue around and about some of these different themes. So it's and and possible to to write more. So just to show what that looks like, just dragging dialogues in or it could have been at dialogues write in new dialogue that will be very YouTube appropriate for all the topics that are referred to in paper, it is a live stream. So, make it most possible like most YouTube live streams do. Okay. Discussions. This is a few different comments that people have uh stated related to the work and I just responded to the um comments in writing in GitHub just cuz it was easier to uh express and and share it. Examples. Uh there's just a few examples and we can revisit this one too. um just this can have a growing number of specific the the linear the linear regression but it's just documents related to examples. So here let's let's do a new now there's up to three parallel chats in cursor. So let's write examples up for the functional taking an output. Maybe in the future these kinds of drags will be less needed. They they help get it right on the first one. Linear regression neural network In examples, please make three markdowns comprehensively for the three main tests output. Have each doc example be comprehensive. Explain why and how it matters. Why and how cerebrum and cases come into play. Show not tell. All the kinds of things pre-addressed that anyone thinking seriously about this would come to wonder initially. Okay, back in the dialogue space. So like that once there's a genre within a a folder even if it's a hyper meta genre like dialogue um it's possible just say okay or add a new dialogue with this theme. Okay so that's the examples

happening. All right. Then there's the insects folder. So, some of this may have uh images that fail to render, but let's look at these in GitHub. Okay. So, here's just zooming in just a couple layers, just a couple prompts deeper into all right, how do we apply cerebrum to entomology? So just like we were previously talking about well the linear regression used as a tool or the predictions of it. So this is kind of uh classic standard into linguistics. What do people talk about when they talk about bugs? And thinking about what what functions, the kinds of things people do and have and and will and would say insects. Taking a first pass at the syntax of what they said, what are those insects being said to do or have done to them or their location, etc. in that method section or in that natural history report. So it's like are we talking about different functional roles different places where in describing that function that natural linguistic case would be used different speculative insect cases mixing the sort of hyper practical discourse analysis uh scientific reproducibility avenue new uh literature annotation um hypercharging mixing with these speculative cases like talking about the ants pheromonal implication. That's kind of an interesting possibility. Now, it might be morphologically silent in English or some other language just as other cases are in other languages. So, it isn't even that the words have to be spelled any differently or pronounced any differently. It could just be a situation where it's understood that the ant as a pheromonal actor or recipient is coming into play. Maybe these are even over meshed with other linguistic cases. So different kinds of topics that come up in ants but different aspects of their activity um that kind of give a special spin or clarity or relevance to different ways to talk about an object under different sorts of constraints. looking at the ways in which this big difference amongst languages of the world the nominative accusative distinguishing models acting from models being acted upon regardless of the actions transitivity this might map to systems with clear initiators and responders so the queen lays an egg irrigative absolutive groups models performing intransitive actions eg moving with models receiving transitive actions accusative distinguishing them from models performing transitive actions eg depositing pheromone this might map to systems where the state change of a model is paramount regardless of whether it caused or received it like whoa whole onlogical assertional frameworks where different languages are like falling in different ways across these two different kinds of grammar systems and maybe there are situations where one or the other has a different feature defined as all the ways in which they're different are ways in which they could be different in function or in fitness in a situation. Okay, that's the least technical of them. here is applying those kinds of case depictions. Like imagine if the case depuff or a prefix or a subscript or a font or

something like steganographic hidden in the word or it's just a reading lens but it's like the projections of the neurons send data give data to the kaix and it's just like so again it's not the only way to use these parts of the brain. It's not associating them with parts of the brain. It's like saying there could be method sections that use these regions and describe these processes using this kind of linguistics. different computational precision parameters and how they are in causal relationship with sort of natural ethological behavioral linguistics about what is happening. Like this is a classic Professor Deborah Gordon point with the space between seeing the nestmate on the ground and then identifying the task it's doing like it is foraging or it's part of foraging and then going from the task identification to the identity assertion like it is a forager but it's like a nestmate that we see nursing is that a nurse is that like is a is a person who plants trees when they're not planting a tree still a planter. That's more classic and to linguistics more from the Vickenstein side. So different ways that stigma G is studied all these kinds of scenarios looking at beehive foraging looking at the algorithms of the different ways that information flows and processes happen relocation. Okay, insect here's more acronyms and puns for more topics. So uh in in the in the kind of last section as we sort of go beyond cerebrum probably today maybe we can use some of these or another one insect cognitive architectures. All right. I was happy to see the harvester ant pogamir max Barbados. Um, so different insect species and ways that they've been studied, even though many or all could be studied for all, but there's just many modeledish organisms that have been studied. Well studied insects, looking at how different brain regions are described of, if not playing functional roles of using different cases. again like are there more mentions in the literature of we injected the die into the projections of the neurons or um is the just like in the different methods sentences of the statistics paper it was like describing the linear regression or the neural network or the palm DP as getting things done to it or doing things or being used with all of those similarly happened for descriptions of insect brain regions. Different classical behavioral types keywords for literature search like let's do a literature search on spatial learning plus himopa pull all those papers download all the papers break them into sentences look at every sentence do entity extraction determine what case different entities are in. So this is kind of like a formal linguistic complement to statistically driven engram learning like classically just taking large bodies of text and then looking okay we could look at the word frequency that's like the one gram we can look at the word co-occurrence the two gram and going on and on and on and that kind of leads to uh you know one thing after another to the attention is all you need which has some kind of engram like basically the context and the

attention mechanisms with a bit more nuance but a lot like that. Okay, more tables just linking up different ways that different functionalities, different you know what what cases would we want to transfer from how we talk about ants in order to be able to say those kinds of things about swarms of robots. Okay, more case studies here is more of a deep dive on insect brain anatomy. So looking at the different sections of the brain in different ways, tables looking at different adaptations. more brain mapping, more connection with species, behavioral functions, neurotransmitters. So, it's like there's again going to the literature and annotating where those kinds of things are said and then that is already just sort of the forensic cerebrum approach. It's just a purely annotative strategy. And then also it it implicitly and explicitly involves sketching out these state spaces even where some of those combinatorics are like grammatically invalid or prohibited for one reason or another. Okay, not all these work. This one doesn't work. Different brain region, different insect research that could be done. Okay. Then there's two folders with a lot of documents. So the first one there is languages. So languages first there's a folder computer. Computer lang. This has uh uh several dozen computer languages. So let's look at go. We'll just look at one computer language and then mainly at natural languages. overviews the code different kinds of use cases annotates the code with different case annotations on different processes. So you can see all of these kinds of abbreviations. So different ways what's one like just on looking at even just one line of code one language might say the data is sent to X from Y or Y sends the data to X or just data X Y. Not to go into this on the computer side but the uh human languages folder is also very rich. There are a variety of languages. Uh the readme is pretty much updated. Let's just look at a few. Esperanto. In the previous stream, we looked at how the Hungarian case system has onto on and from a surface. So, sort of perfect for different functionalities of interfaces and and boundaries. interlagic Russian English goes into more detail just grammatically and cognitively on on some common first and second language learning combinations of Russian and English and like why certain failures of language are common cognitively and grammatically. Sanskrit Tibetan. So, Esperanto overviews the case just like kind of a more like a grammar textbook, like a really fast summarized grammar textbook. Sanskrit went into a bit more uh detail here. Just several times it would add a couple hundred lines. I'd say awesome add more interesting, funny, useful, meaningful sections. So different case um information different ways that a model like an NLP model could be used according to those cases. grammatical number one, two, many where some languages might only have one and many but you know it could be helpful. Are we talking about a single agent? Are we talking

about the diadic communication setting or are we talking about the more you know three three is a party different syntactic speech features so this could relate to if we have a given model and just we're going to be using Python or subscript or special characters or prefix it's like whatever um morphology setting you're in with the model there might be different kinds of of um certain things might be easier to express or others or might have different consequences. All right. Then just kept on going and just say add more. So lots of more um more connection with temporal and modal systems. So like this starts to get into the beyond cerebrum stuff that we'll do in a few minutes. But like there are other parts of speech and other linguistic techniques other than case even though cerebrum has focused heavily on case in these kind of two stream/ early versions of the paper. But of course there's a whole bevy of conditionalities and moods and all these different kinds of things which like could be used to describe even just statistical models like while this is happening if that happens then this will be happening at the same time as viewed from this perspective. So different languages have different natural expressivities, ways of modifying how things look and seem. Some languages modify certain things for sex, others don't. Some modify for number differently. They um of course have different heritages and and connotations and and different concepts that are sort of already there. And then also they they all have different and and including created contrived semantic systems as well. They have some commonalities and differences. So for certain kinds of computational semantics there may be some very small programs or some very secure ways to do this or that and then Tibetan. Let's just see trust systems, evidentiality markers, evidential systems, direct sensory evidence, general knowledge or inference. personal experience or witness with suffixes, direct, indirect, witnessed, hearsay, and inference. Also think about legal um ontologies and and different ways that this kind of relates there. So that's the languages folder. There's all the computer languages and then I mean not every single one but that is where all the computer languages are and the natural languages. All right, last folder to look at is the speculative design. So just to show uh kind of similarly here. Okay, just to quickly return to the um examples, let's push GitHub, go to the examples, and then uh look at the three new that were written just now. POM DP, linear regression, neural network. So introduction laying out what are all these different cases about how do they apply to the palm DP what does the demonstration do linear regression background on linear regression laying out what is different and why it matters. and the neural network case similarly. Okay. In the speculative design folder, we will add one more document. Okay. Dragged in speculative design. In this folder, write the most vast vastly layered comprehensive synopticon of the just going to drag

in the or I'll just drag in the single markdown output. assembled paper. Write the most and do it absurdly and information densely in the style of something that strangely would make a lot of sense to see visually on a live stream about this topic yet be essentially incoherent in the linear reading. Conveying all the concepts totally The user wants me to create a new file Okay, we're looking at the chain of thought of Gemini. So called chain of thought. They want me to create a new file, assembled paper.md. Not really. I didn't say that title, but let's go with it. It should be a synopticon. It should be visually in interesting but incoherent. plan. List the files. Read the files. Synthesize the content. Create the file. How to make it absurdly layered. All these different strategies. How to make it incoherent linear reading but make sense visually. How to convey all the concepts. File list. Then it lists out the files in speculative design. It's like, you know, it's not exactly what I thought I wanted to know or see, but but it's starting to get to a very interesting place. For each snippet, randomly or pseudo randomly, maybe based on file order, intersperse headings, bold terms, list items, and short sentences, phrase into the main synopticon string. Use heavy markdown formatting. Prioritize density and visual chaos over linear coherence. Oh, brother. Ensure terms, concepts from all files are included. Okay. still going. This is just the beginning. Incorporating elements. It's like this is doing quite many cycles of thinking, reading, doing. Lots of tokens being generated here. But meanwhile, let's let's let it cook. And uh there's a few that are pretty fun here. So Okay. So, here's a Daisy Dolly Rimple um satire uh free speech usage as they all are where it's a sort of early 1900s detective story and the first part is is just story. Then it goes into this truly delightful sequence of uh looking at the first off speculating some some ancient origins looking at the different I'll turn off dark looking at different hypotheses sort of like this like mystery science theater 3000 reading of a Dolly Rimple situation talking about how in their case in their situation in their case in their cases how different processes happened different context shifting different processes that different ways that that then it kind of comes around at the very end to uh highlight the unique values of the different languages and then I believe they they call it a night and have some Hungarian dinner. Okay, this was one that I just said make a new speculative design document. Go somewhere that's that's never been done or something, you know, glitch ontology and error-based learning. So it's like kind of glitch pop which has like a visual and a auditory. Uh so it's kind of a multimodal glitch but it also comes up in programming and it's kind of a digitized format. So like different ways that glitchbased methods we can fix some of that. New tab glitch ontology fix and improve all the mermaid diagrams. This is probably the longest um Gemini sequence. Oh, okay. So, it ended there.

Continue to refine the synopicon and read all papers and then save the final full file. Could be slight differences in mermaid rendering. Um, but cool idea. Glitch-based ontologies like ontologies of errors for describing systems that have certain relationship with error. Okay, this is a very long one but a but an extremely worthwhile one looking at William Blake. So it started with just this bullet point type stuff now starts getting into Blakeian engineering like the four-fold vision discussed elsewhere with single vision being like leading up to metacognitive integrations. So it's kind of like this is not dissimilar than a three or four layer hierarchical model like thought seeds or sanded smith type model and then it just shows like okay which model in that four-fold vision paradigm going into detail like this is kind of like the retinal layer with the convolutional neural networks dropout layers that kind of stuff because it's like the sensory so it's like multi-scale higher hierarchical architectures, possible self-similarity of the two, three, and four-fold visions in terms of their capacities going into the zoas looking at cognitive functions of the integrated albon and the different kinds of functionalities fractal self-similarity with eternity in an hour. How would that really get implemented? um different cosmological parallels. What would really be different with a Blake inspired cerebrum using AI? Well, single objective, contrary, balancing, multi-objective like to target the m the creation and maintenance of tension or paradox architecture detailed architecture layers. So it's kind of like there's the does this accurately describe what Tharmmas is in the Blakeian ontology. That's kind of like a separate question that's interpreting this just as like structured humanities research and just is that uh mer of the different places this box could be pointing or connected to. Does that align? And and surely there's multiple uh takes even for one character and then it's like what are the functional attributes of this kind of architecture. So what is that transfer path from loose or ambiguous or or multiple or even contradictory poetic description and basically all of these kinds of information processes generation reality layer the lua generative layer that's the two-fold vision three-fold vision golanuza creative layer and then the Eden integration Okay, there's just too many funny things. This this was a funny one. The eternity sunrise development. So 2025, this very year implement the core zoa's architecture begin contrary processing research. So little different spin than predictive processing. Okay. Contrary processing research. Hm. But that's that sort of unity is plural at minimum two research agenda. But then that's also the paradox. Well, but then isn't the single objective unity is plural at minimum two? So how how to hold the one and the two? That's like the single and the two-fold vision. Then over the years leading to 2029 eternity sunrise self-evolving cognitive architecture begin Albon integration more sections

then really nice process diagrams super dense looking at at different and being able to do programmatic generation archetypal description of logistics for perception action cycles. So this is kind of like a whole cognitive sequence. It's not just saying well there's a for loop that gets an observation and then uh puts out an action or it's like an observation comes this parallel processing like marriage of heaven and hell status occurs certain sequences happen then there's a back and forth here and then this all etc. It could be make a dynamic control flow in the marriage of contraries. Let's do make a vast comprehensive dynamic control flow with massive level of detail of process diagrams. Something com completely capturing the spirit of James Joyce. All the most chaotic Riverrunesque stuff. Also do the same in another design file for cerebrum plus infinite jest. Okay. Wow. Still plugging on that. Those are almost certainly the two very longest um executions I've ever seen from Gemini for for for the sheer number of uh steps. Okay. Then the last one before we push those updates look at the new ones. This was uh as per the uh request of a friend on John D who lived from 1527 to 608. So I was not extremely familiar with D but this was a really interesting approach that brought together those process diagrams um different kinds of cgraphy spatial computing time cryptography secure communications. So it's kind of like John D's work transferring onto what what was said in that work and what has been said about that work onto the specification of cryptographic or of like esoteric semantic components. Correspondence tables, active inference topics, category theory terms, different ciphers. We can do a whole Okay, that was okay. That was it. Did the full assembled paper. Make vast speculative design document on using cerebrum or cryptographic design, syntax, semantics, cognitive narrative. This one halted early. Okay, let's look at the assembled Okay. Synopticon. This was prompted again to be incoherent to read out loud but to give a lot of information when visually looked at Okay, let's look at those the joy and infinite chest. Then we'll look at the live chat. If anyone has an idea or question, then go into the sort of beyond. So here it didn't make the file, but that's what's so fun about cursor. Put that and improvements and additions into a file with the right name. And also this chat is getting long. So that's why there's this start a new chat for better results down there. Okay, push these Okay. Joyce and Jest. Okay. Infinite chest. First one doesn't work. Basian funhouse mirrors. This sounds clean. It rarely is. Okay. And the joist. I don't think I can evaluate the details, but you know, it's cool. Okay. And there's the cryptographic. That's what you know again it just it's the concept. It's it's the possibility that these spaces can be defined up front or on the fly and they can be passed to the right other agents or people or just generated from scratch or in the future in a better way. Okay. Okay. So

the main piece to sort of leave a little snippet of again in this last part it's going to be just a few uh what's next in our path and then if anybody has a idea or a question live chat that they can write it. So, let's go into a whole new folder beyond cerebrum. Beyond spec. So given all that information on cerebrum and all it can do with grammatical case. Let's zoom out and think beyond Let's come up with insects or other invert related acronyms to conveying absolute total scope of what can be considered a quote full linguistic intelligence method. whereby all the parts of language, verb, noun, all parts of speech, moods, etc. are firstly formalized into the kind operations which flexibly can and do apply to eg linear regression neural network basian graphical model. Then in that case with full formalization we are only in the tip of the iceberg in terms of xeno and exol linguistics. Firstly come to totally understand that then spec out completely this quote beyond cerebrum effort. Okay. planning a response. Pormica H. That's a that has a nice ring to it. Myriad, Mantis, Apis, Locust, Vespa, Arachnne, Silk. All good. Okay. Formica Locust Arachnne. Let's go with Formica. Okay. First idea is the best one. continue just even just pre pre-prompt pre-writing what I I believe or expect I will want to say soon or next with all needed files functions underpin Ed by same functions described in core beyond spec document which is always growing and improving iteratively such as forensic work describing the lexical environment of a model in time in place in use. All modern machine learning cutting edge questions interpretability scaling safety pure enmological linguistic pragmatism. Wow. Already jumping to the tools that you know it could be a it could be a uh Wow. I've never seen a chain of thought that looked like this. Create directory structure. Okay. Source a source folder within. Then update the spec. Create the init scripts. Add a readme. Okay. Now this is the what? This is the part I've never seen. Constraint checklist and confidence. Follow cursor rules. Yes. Here's the cursor rules. One funny prompt in another setting was like, "Write yourself a prompt that will lead you to generate the prompt that will help you generate the greatest, most accurate, relevant, possible cursor rules file set. Okay. So it's like this is kind of where Q kids and catechisms come into play which is just developing these sets of questions. So it's like oh yeah that's the Q78 or like oh you you could you could just use the use the Q600 ask all those figured out you know. Okay placeholder for lexical environment. So for this one we can uh download we could scrape uh papers or scrape websites and then use anything from rulebased methods to existing software packages to LLM API calls to say okay given what we know about cerebrum and whatever this form or whatever system is given that what are the entities what cases are they Yes, given the structure now then comprehensively continue so that all docs formal specs of the parenthesy more than

language formal type theory category calculus and basian cognitive functions are 100% 100%. It's not sort of where I hope to and will go over the coming little bit uh with anyone's open source participation if they're if they're interested listening along this far. Let's read that doc. Okay. Cerebrum provides a powerful framework for integrating case-based reasoning and basian representations, focusing significantly on the role of grammatical case in structuring meaning and reasoning. While valuable, case grammar represents only one facet of the complex system that is human or potentially non-human language. To achieve a truly comprehensive computational understanding of linguistic intelligence, we must expand our scope significantly. New chat tab. Speculative design. looking past even future horizons. Write a vast comprehensive speculative design doc with valid mermaid diagrams and tabular intelligence about what gle maximalism really means. parenthesy in group discussion only. Okay. Okay. Define base types for fundamental linguistic units. Phony, morphe, lexical item, syntactic constituent, semantic concept, pragmatic concept, composite types, dependent types, connections to formal grammar frameworks. Laying out those types research challenges relation. Okay, but just continue. I'll start a new chat for better results. Continue comprehensively with developing Have all needed documents in fractal docs folder. Have all source code and tests. Start running and developing the full simple modular test suite now. Okay. Okay. Question from the chat. What kind of predictions might you have about case users? across disciplines, experience in a field or experience with the language. What would it reveal if case use was different across groups? Okay. And another question live chat. Can we me can we mechanically deform the cranium to 10x the cerebrum? Um, probably. But you probably couldn't get it back, Tucker. But I think I think with a 3D printer and a shipping container, it could be possible. What would reveal if case use was different across groups? Yeah, there'll be so many interesting topics like the uh discovered slashinvented uh question of case to what extent are two on at from of an object fundamental aspects of semantics. Are there other fundamental aspects of semantics that we get through or get around in human language in other ways? I think that at the very least a first step, not the only first step, um not not the not the last step, but a first step is this kind of beyond cerebrum approach where it's like first get the full interoperable category theoretic calculus type, you know, algebra, etc., etc., etc. for all the languages that we know about. partially to scaffold and improve the fluency of people and LLMs, partially to understand well what are all the state spaces and then if we find that there's something where there's always exactly three in every language even if they were all the same we could still say like well what if it was four or maybe we'll find that

there's um one two and many for marking numbers in one language and another one it's one two three and many another one it's just one and many or another one doesn't have that distinction it's any of those even just seeing two examples is kind of like the arc challenge. It's like you see that uh this could have been this way or that way and then there's a whole state space that it could be. So there could there could be a uh uh special numerosity for multi- agent models where it's like there's a subscript for how many agents there are. But it can it can have, you know, 50 subscripts coming out of every corner, different fonts, all these different kinds of modifications. And that's just typographically, let alone how it's spoken or how it gets rendered computationally where it doesn't even need to be limited by like how many suffixes can you add or or how many um subscripts would make sense in uh your your word processor. It's about these like higher order modifications. So when people say I'm talking about sending the data to the affective active inference models API that could be something that gets fully fleshed out maybe static analysis on certain components but certainly for scientific integrity full reproducibility understand what is that category of models and so that is the curious language of the incident which is for the incident which motivates languaging. What is the curious languages that let that semantics get communicated? Okay, some more great questions from Tucker. We will make documents. Okay. It seems like human language taking a survey of different semantics and grammarss around the world doesn't need case and that even semantic density is not specifically correlated to case versus no case structure. Yeah, it's it's highly possible that um by kind of sandpapering the language of the exoteric markers of case like English for the most part it it it could be associated causally or just correlationally with an increased or decreased information density. So consider different ways like per phone. Um whereas like if you're doing stem modifying like a Slavic language then maybe there's only so many modifications that you can overlap. If you're doing suffixing then maybe you can have more but you know suffixing might lead to very long symbol density. Then again if there's a certain number of symbols that are needed for the subscripting you know just so there it's I I don't think there's a single uh answer especially in the absence of empirical evidence. uh in our in our uh hands and mandibles. But we could say write a speculative design dissertation summary of all the situations where human languages with and without different case systems would and would not have different information densities. designs considered at syntax, semantic, narrative, cognitive levels. Okay. Fill in all placeholder methods. Start to run log confirm the test suite formica. Okay. Gle maximalism. Oh wow. Being a little bit um recalcitrant to write this document despite its absolutely benign cellular nature. Write that and

improve that into a Okay. Okay. Tucker wrote, "Why do so many ancient languages send? I don't know what us is." Usakri You Scotty. Oh, thank you for that. That we should add this to languages. Oh, it is there as Basque Uscara. Why do they have no chill? I don't know. Now fully write out that dissertation. Tons of tabular formatted crystal good stuff. Parenthesy information. Vast technical mermaid diagrams. Unique explanations and predictions. Unknown and unknown to linguists in 2025. Wow. Still still dragging heels on the component. Maybe switch to Claude. Oh, great question. Great question. Let's ask perplexity. Okay. Tucker wrote, "Are any modern languages in widespread use gaining in case complexity?" Let's do the perplexity research. case evolution today MD. Okay, that's interpreting that. Okay. research and fully empirically address this question from my good friend. The answer must ring in his ears like gold and silver. And he is a refined listener. He expects only the finest information. Okay. Meanwhile, Perplexity, let's do the slower or the faster mode of Plexity. I don't know. Yeah, that would be uh this is kind of a uh one one approach that could be used beast which is basian evolutionary analysis by sampling tree. Let's write out this document. This is going to be beast case analysis. Okay, I'm just going to copy in just one paragraph about beast. Oh, we have so much good stuff happening though. Okay, that one will finish soon. That one will finish soon. Let's just see where he got with the development. Yeah, making good progress on Formica. Okay, with this description of beast, write up a fully technical NSF posttock proposal for using cerebrum and formica parenthesis. Please lift them up if you need to do ancestral state basian phenotypic reconstruction of ancient linguistic elements. We are going way into the past. We will use generative AI as needed for creating create creating fluent text, audio and visual media. Given the explicit probabilistic models and empirical data, we will be compiling and inferring on and contemporary languages. Okay. Cleo maximalism. Oh, it's okay. It says it's for internal discussion only. I feel like I could share it here though. But we'll reset chat. Go back to Gemini. Write it out fully. The full posttock application. Thank you. Okay. Gle archetypes. Oh, this will look better on. Okay. Test infrastructure. Okay. We stop the agent after 25 calls. Continue. Look at the gle tests developing writing the that posttock application. Okay, but to Tucker's question, most linguists agree that human languages tend to simplify their case systems over time rather than develop more complex ones. Of course, just like in genomic and uh phenotypic settings. It's like wait, but then how was there a more complex one? The historical trend seen in Indo-Uropean languages has been towards case reduction, not expansion. Latin, old English, old East Slavic and Sanskrit. All of them have dropped in cases. Are we buying the dip with respect to case

number? I don't know. Maybe it could go down further. Not not financial advice, not linguistic advice. Okay. Counter evidence. Virtually no documented examples of modern widely used languages have developed more complex systems as on 11 wrote in languages cases usually die out. It's a known process. Why would they appear instead of disappearing? Right? Why would they appear instead of disappearing? Absolutely. Perplexity. Let's see. There's no strong evidence that a modern language but but but could could a language with more cases gain more popularity. Are any languages gain case? But English is right now rare but possible. Oh, they debate about Mandarin Chinese. Okay. Deeper research. Okay. But you now you you know what I am going to ask next. If that case reduction is such a slam dunk that it is pervasive today. Give me the complete parenthesis and esoteric account of how those complex many systems ever even came to be. Okay, gal cells. Okay, so centering the role of the GA and then there's the neuronal processing layer, but it's like we we don't really talk about those in GIA science. It's more about the kind of multiple types of GA micro gal criticism. Okay. So that's kind of like feedback. Okay. centering the role of the oligodendrite between neurons using formica in the sort of neural semantics Gal. Okay. The emergence of elaborate case paradigms and languages such as protoindoeuropean pi Sanskrit and finick languages involve layered grammaticalization processes and socio factors. The pi case system. So this is the one that is mostly used in cerebrum. The system likely arose through spatial and semantic differentiation but I repeat myself. Analogical extension case markers originally tied to specific semantic roles. Instrumental for tools expanded to abstract syntactic functions. Phonological reinforcement. Sound changes eroded distinctions between stems prompting speakers to innovate new markers to avoid ambiguity. Stable case systems. Oh, some languages did have expanded case complexity. I mean, like maybe Hungarian or some of these other ones. The dative plural um persisted due to its role in marking indirect objects. Yeah, interesting. But a little little but but that's exactly that's exactly what we can analyze with this postto application that we're writing just like you could use beast to look at the genomic or phenotypic patterns. Okay. Run the full Formica test suite. Develop and improve the Formica tests, methods, visualizations, animations, documentation. Okay. Okay. Vastly expand the NSF postto Add comprehensive tables, specific explanations and in huge valid mermaid diagrams. Okay, still plugging away on Formica. Okay, we're going to get this Postto application done, read it on GitHub, see if anyone else um has any questions or ideas in the live chat, and then call it a day. Okay. Yes. develop everything fully run and improve that test suite which yields full computations and visualizations. Okay. Right. Just save that. a few improvements on this

kind of stuff with like the need to reapply tools and uh better situational awareness for the codebase reading folder structures kind of doing better in uh that could be partly in cursor partly uh other other approaches. All right. Speculative design beast. Okay. Reconstructing ancient linguistic phenotypes using integrated basian modeling and casebased reasoning enhanced by generative AI. Okay. intellectual merit pioneering the use of generative AI specific generative AI model name eg GBT4 tacoron I'm sorry it is an end to end text to speech tacoron broader impacts Okay. Oh, that's going to cost you foil. Yes. Lexicon swadesh lists. Perfect. It's like there's just so much to learn. A lot of it is out there. A Swadesh list is a list of forms and concepts which all languages without exception have terms for. Okay. New file just dropped. Swadesh list. Yes. Run the test suite and add tests that show off and visualize for new chat. Write a swadesh list for cerebrum and formica as language. Okay, thank you Tucker. Then write another file meta swadesh list markdown both maximum length and accuracy. Yeah, cursor writing these terminal commands, running the pi test by itself. Super big speed up, super effective. Okay, hyper swadesh. That next. Morris suadesh. He started with a list of 215 meanings. falsely introduced as a list of 225 meanings in the paper due to a spelling error. Yeah, hate to see that happen. Whoa. Wow. Great topic to have. Okay, this is the swadesh minimal minimal terminology. Who, what, what, where, when all could be associated further and furthermore the swadesh list include all swadesh terms and add case annotation. Okay. Then the meta meta swadesh. Oh then. Okay. Clearly there has been a kind of misunderstanding. A meta-wadesh list is like a swad list for such lists. A list of all the criterion you'd use in the production or inference productions or inference of a Suadash list coming to a file near you. Okay, Formica is up and running. Add a visualization script for Formica. Ensure that running The test suite yields copious accurate visualizations and do gif. Okay. You're doing okay with the sweat dashless. Okay, let's look at this. What the the the the first the layer one Swadesh list the L1. Yes, perfect. Like this is the basis of what we need for Formica, which is what what are the what are the 200 or so concepts that we need in Formica for total ant colony modeling? What are their parts of speech here? Nest core their forage zone. So it's like home and away like over here over there. Many many mandibles. Other alien scent. One, two, three, four, five. Yeah, I guess you don't even need six. Maybe you call that just like le legish or something. Okay. Different different um different creatures in the nest. Okay. body parts, actions, natural colors, adjectives, directions. Great. Okay. Okay, now we're on the right page with the meta swadesh list. Now write that meta swad list fully. 200 concepts or so please. Yeah, run the test. Suite confirm and log where visualizations are made. Okay, I

understand you want a significantly more detailed and exhaustive breakdown of criteria, principles and considerations related to Okay. Our next album MD. Okay. New chat. Prediluvian Super Swadesh is the name of our next album. Write out all the song names and lyrics and in call outs describe the that sound. Okay. Meta swadesh list. We're there. It really did go up to It It really did go up to 207. See also external resources on Swadesh lists, Lexico statistics and GLTO chronology. Write another comprehensive document on glto chronology. Okay. Okay. 27 tests. Looks like beyond cerebrum is coming together. Let's see where it's saving things to. Yes. Run the script and then run a terminal command to confirm where the visualization saved to. Glauto chronology is a controversial method within historical linguistics that aims to estimate the time depth since two or more related languages separate from a common ancestor. It attempts to do this by quantifying the degree of laxore placement within a specific subset of core vocabulary. Yeah, I think basian methods could help a lot there. Okay. our album. We're gonna look at this album on GitHub. All right, this is going to be this will be the last piece. I think people get it. It's possible to um go beyond cerebrum in like a 20 minute period. Okay, we'll close with this. The pre-diluvian super swadesh. Silt speaks first. Primordial murmur. Obsidian shards on the shore. The cgraphers's dream. That's an instrumental. Occasional occasional sonar pings or distorted radioatic hint at failed surges and forgotten geographies. Whispers from the ziggurat's core arc construct gopherwood and pitch 40 days delusion. Dissolution ararat landing new soil old son. Well, hope this has been a uh interesting stream for people. Uh, it's all Creative Commons BY NCD. So, if you want to work on it, feel free to make contributions to the repo. If you want to collaborate on a future part of the Formicus stack, there's uh there's always a always a way and we should be we should be on our way. Okay, that's it. Looking forward to people's comments. Thank you, Gia Maximalist and Jeff and Tucker for the live chats. Okay, bye.